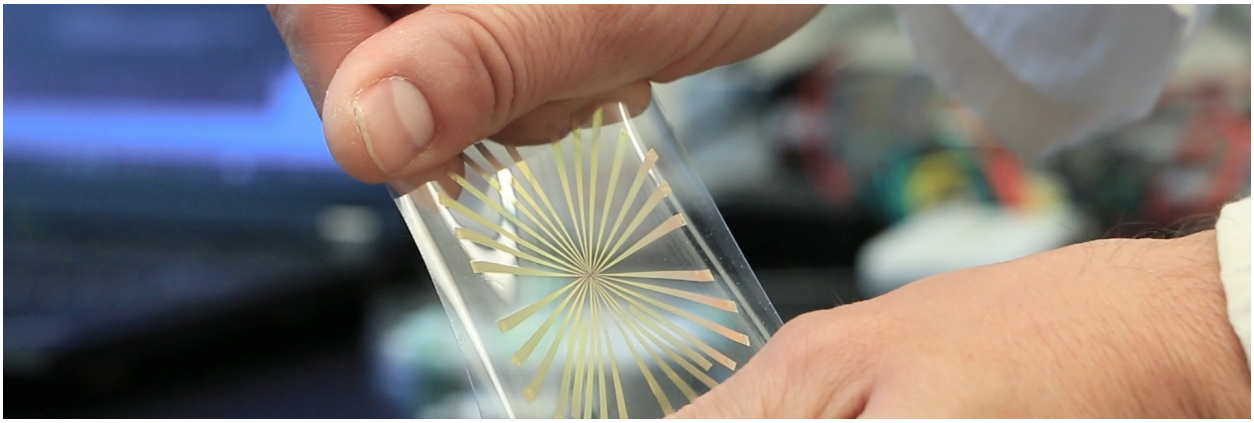


BME 202: Control and Communication in the Nervous System

Fall 2027 | Units: _____

Instructor

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Professor of Biomedical Engineering, Biological Sciences & Neurobiology
jlopez@usc.edu
Office: _____
Office Hours: _____

Course Times & Locations

Component	Days & Times	Location
Lecture	Monday / Wednesday / Friday; 80 minutes Time: _____	_____
Laboratory	Separate hands-on sections beginning Week 1 Days and times: _____	_____
Exams (4 total)	Exams 1–3: Mondays, 6:00–7:20 PM September 20, October 18, November 8 Exam 4: Wednesday, December 8, 4:00 PM	_____

Course Description

BME 202 examines how the nervous system generates electrical signals, communicates information, and controls behavior. The course introduces the structure and function of neural systems through the biological mechanisms and engineering principles that explain their operation, from ion channels and membrane potentials to sensory processing, movement, and neural interfaces.

The course is organized around four themes: the electrical foundations of neural signaling; excitability and communication between neurons; neural coding, circuits, and biological control; and sensory systems and neural technologies. Across these themes, we connect experiments with mathematical models and circuit representations, examining how measurements constrain

a model and how changes in a biological system affect its behavior and the design of biomedical technologies.

Lectures combine explanations of neural mechanisms with instructor-led worked examples, calculations, simulations, and interpretation of experimental data. Approximately half of lecture time develops the concepts and half applies them, with examples integrated throughout the explanation. Separately scheduled laboratories begin in the first week and provide hands-on experience building circuits, operating instruments, recording signals, and testing the models introduced in class.

Prerequisites

MATH 126 or MATH 129. No previous biology or neuroscience coursework is assumed. The course builds on calculus and introduces the circuit, differential-equation, and computational methods needed to study neural systems. Prior completion of MATH/NEUR 265 is not required.

Course Materials

Reference textbook (optional): Kandel, E. R., Koester, J. D., Mack, S. H., and Siegelbaum, S. A., editors. Principles of Neural Science (6th ed.). McGraw Hill, 2021. The textbook provides additional biological background and explanations of neural mechanisms.

Assigned readings and research articles support the mechanisms, models, and biomedical applications discussed in class. Readings identified as required are part of course preparation; the reference textbook is optional unless a specific selection is assigned.

Lecture material and PowerPoint slides are the primary source of instruction, and exams are based on the material taught in lecture. Worked examples develop the calculations and reasoning used to analyze neural systems, while the readings provide context for the biological mechanisms and experiments behind them.

Although this is a lower-division course, the material builds quickly and requires consistent practice. Understanding an equation includes knowing what its terms represent, when its assumptions are reasonable, and how its prediction relates to a biological measurement. Regular review of the lecture examples and steady work on problems and models will be important throughout the semester.

Required equipment: A laptop capable of running Python and Jupyter notebooks, a scientific calculator, and access to the course site. Python is free, and MATLAB may be used where university access is available. Prior programming experience is not required; computing examples and templates support the biological and engineering work. Bench equipment and recording systems are supplied in lab. Contact me if access to a suitable computer is a concern.

Course Objectives

- Explain how ion channels, membrane properties, and synapses produce and transmit neural signals.

- Use circuit models, differential equations, and numerical methods to predict neural responses.
- Interpret neural measurements and evaluate the assumptions and limitations of a model.
- Connect neural coding, sensory processing, and motor control with biomedical technologies and nervous system dysfunction.
- Build and test experimental systems, collect measurements, and communicate engineering conclusions supported by the data.

Teaching Assistants

This course has no co-instructor. Questions about lab logistics, reports, software, and assignment expectations should go to your assigned TA first. Concerns about scores or grading feedback, including requests for regrading, should be sent directly to me at jlopez@usc.edu.

Role	Name	Email	Office Hours / Location
Head TA	_____	_____	_____
Lab TA	_____	_____	_____
Exam Proctor	_____	_____	_____

Grading

This course has two components, lecture and laboratory. Each is scored out of 100 percent. Your final grade is 50 percent lecture and 50 percent laboratory. **You must earn at least 60 percent in BOTH components to pass the course.** A strong lecture score does not rescue a failing lab score, and the reverse is also true.

Lecture Component (100%)

Assessment	Weight of Lecture Grade	Points
Exam 1 (Lectures 1–10)	20%	120
Exam 2 (Lectures 11–20)	20%	120
Exam 3 (Lectures 21–30)	20%	120
Exam 4 (Lectures 31–41)	20%	120
Homework Sets	10%	60 (scaled)
Neural Simulation Project	10%	60
Total	100%	600

Laboratory Component (100%)

Assessment	Weight of Lab Grade
Lab Reports (8 reports, all counted)	80%
Dry-Lab Workshop Checkpoints (4 at 2.5% each)	10%

Assessment	Weight of Lab Grade
Laboratory Attendance	10%
Total	100%

Dr. J's Grading Philosophy and Curve

Biomedical engineering requires you to connect biological mechanisms with quantitative reasoning. Memorizing an equation or reproducing a worked solution is not sufficient; you must be able to explain the model, choose an appropriate approach, carry out the calculation, and interpret what the result means for the system being studied.

Exams and laboratory assignments reflect these expectations. Some assessments will be difficult, and raw percentages may be lower than you are accustomed to. Grades reflect demonstrated understanding, scientific accuracy, and the quality of your reasoning. Consistent effort and participation are important, but they do not substitute for meeting the course's academic standards.

If the final class average, calculated using the weighted lecture and laboratory grades before curving, is below 70%, a curve will be applied at the end of the course. If the class average is 70% or higher, no curve will be applied. The size of any adjustment will depend on overall class performance. A curve accounts for assessment difficulty while maintaining expectations for mastery; it does not guarantee a particular letter grade or a passing grade. Use early assessment results to identify gaps in your understanding, review feedback, and seek help promptly. Do not rely on a possible curve to compensate for missing work or consistently weak performance. You must still earn at least 60% in both the lecture and laboratory components to pass the course.

Grading Scale After Curving (changes year to year)

The percentage ranges below are approximate estimates, not fixed or guaranteed cutoffs. The grading scale may change from year to year, and final cutoffs for this semester may differ based on overall class performance and any curve applied under the policy above. Use this table as a general guide when estimating your grade.

Letter Grade	Approximate Percentage Range
A	90-100%
A-	85-89.9%
B+	80-84.9%
B	75-79.9%
B-	70-74.9%
C+	65-69.9%
C	60-64.9%
C-	55-59.9%
D	45-54.9%
F	Below 45%

Final grades are not rounded up. Letter grades will be assigned strictly according to the finalized cutoffs, which will be announced before final grades are assigned. Regardless of any adjustment to the scale, you must earn at least 60% in both the lecture and laboratory components to pass the course.

Exams

There are four exams, each worth 120 points. Exam 1 covers Lectures 1–10, Exam 2 covers Lectures 11–20, Exam 3 covers Lectures 21–30, and Exam 4 covers Lectures 31–41. Exams are not cumulative, although later material uses earlier methods. Exams 1–3 take place on Mondays, September 20, October 18, and November 8, from 6:00–7:20 PM in the separate exam session. Exam 4 is Wednesday, December 8, at 4:00 PM; its end time and all exam rooms will be announced.

- **Number of questions:** Each exam contains 8 multi-part problems totaling 120 points. Problems combine explanations of neural mechanisms with calculations, interpretation of experimental data, and evaluation of models or biomedical applications.
- **Exam versions:** Each exam comes in three versions (V1, V2, and V3). The three versions contain the same problems, with their order varied within each lecture section.
- **Question order:** Exam questions are ordered by lecture. Questions on Lecture 1 come first, followed by questions on Lecture 2, and so on through the last lecture covered by that exam.
- **Room and version assignment:** Before each exam, a randomized algorithm assigns students to the reserved exam room or rooms and to one of three versions (V1, V2, or V3). Assignments are posted on the course site before the exam, and requests to change them will not be accepted. Approved accommodations are implemented through the university process.
- **Exam format:** In-person paper exams emphasizing written explanations and engineering problem solving. Problems may ask you to construct or interpret a circuit model, work through a quantitative calculation, analyze a recorded response, or explain how a change in a biological parameter affects system behavior. The scope and depth follow the lecture explanations and instructor-led examples.
- **Expectations for answers:** Show the reasoning behind your approach, define quantities clearly, use appropriate units, and interpret the result in biological and engineering terms. A numerical answer without a defensible method or explanation is not sufficient.
- **Missed exams:** Make-up exams are not given. If you miss one exam, a zero will initially be recorded. If the absence is approved for a valid, documented medical or official reason, that zero will be replaced at the end of the semester by the average of your other three exam scores. This replacement uses your other exam scores, not your final

course grade. An undocumented or unapproved absence will remain a zero. Approved accommodations, religious observance, and university-required alternatives take precedence.

- Bring a photo ID, pencils, and an approved non-programmable scientific calculator. Phones, smartwatches, and other unauthorized devices or materials are not permitted.
- **Exams are closed-note unless an alternative is authorized for the class before the exam. Any permitted reference sheet must follow the announced limits; handwritten and printed materials are not interchangeable unless explicitly allowed.**

New lecture content is introduced on exam days because exams meet in a separate session. Regular MWF lectures meet normally, and material introduced after an exam's coverage block is assessed on the next relevant exam.

Homework Sets (10% of Lecture Grade)

Homework sets give you regular practice with the calculations, models, and biological reasoning developed in lecture, including problems that extend the worked examples to unfamiliar conditions. Sets are assigned during weeks without university holidays, with the eleven due dates listed in the calendar. All sets count, and the average homework percentage is converted to 60 points within the lecture component.

Lecture Project: Neural Simulation (10% of Lecture Grade)

Each student develops an individual simulation of a neuron or neural system studied in the course, connecting its biological mechanism with a model that can be tested against measured or published responses. The project develops across the semester through the laboratory workshops and is worth 60 points.

Neuron and Neural Systems Simulation

The simulation project brings together the course's biological and engineering foundations through a working model implemented in Python, MATLAB, or an equivalent tool. Your model should reproduce a meaningful feature of neural behavior and examine how that behavior changes when the system's parameters or inputs change. The final submission consists of the runnable model, figures comparing predictions with experimental evidence, and explanations within the code or notebook, with a demonstration during the Week 15 workshop.

Project Milestones

Milestone	Due
Simulation proposal: neural system, model question, and initial evidence	Fri Sep 24, 11:59 PM
Progress check: preliminary model and comparison with data	Fri Oct 22, 11:59 PM
Final runnable simulation, figures, and code/notebook explanations	Fri Dec 3, 11:59 PM

Laboratory Component

The laboratory is a required, graded component that meets each week outside lecture, except during university closures. Eight hands-on experiments provide experience building and testing physical systems, using electrodes and sensors, and recording measurements. The remaining meetings are dry-lab workshops for data analysis, simulation development, and completion of laboratory work, connecting the physical experiments with the models taught in lecture.

Laboratory safety and access: Safety and instrument training are part of the first lab and must be completed before using the equipment. Surface EMG uses approved, isolated recording hardware and voluntary participation, with an equivalent bench-based option available without penalty. Work connected to a participant uses only the approved isolated system; bench sources and improvised stimulation circuits must never be connected to a person. Course activities do not authorize outside recruitment or diagnostic testing.

- **Lab Reports (8 reports, 10% each, 80% total).** Each hands-on experiment is followed by a 4–6-page laboratory report explaining the question, prediction, methods, results, and interpretation, with references supporting the discussion. Figures and analysis files connect the measurements with the conclusions drawn from the experiment. All eight reports count toward the laboratory grade; none are dropped.
- **Dry-Lab Workshops (4 assessed checkpoints, 2.5% each, 10% total).** Workshops provide time to refine the circuits and prototypes developed in lab, analyze experimental measurements, and work through questions that arise in laboratory reports and simulations. The assessed checkpoints follow the development of these skills across the semester, while the other workshops support ongoing work and the final simulation demonstration.
- **Laboratory Attendance (10%).** Attendance is required at the scheduled hands-on labs and dry-lab workshops, with one absence excused without an attendance penalty. After that allowance, each additional unexcused absence deducts 5 percentage points from the overall laboratory grade, even when the total deduction exceeds the attendance component. The deduction is applied once per absence, and missed reports or workshop work remain subject to their own requirements.
- **Make-Up Laboratory.** One make-up day is scheduled for Tuesday, November 23, before Thanksgiving closure. You may complete only one missed hands-on experiment during this session, with arrangements made in advance so the appropriate equipment is available. The associated report is due December 3; completing the experiment does not automatically remove a penalty for an unexcused absence. The optional make-up session does not create an additional attendance requirement.

Lab Schedule and Due Dates

Lab / Workshop	Week	Topic	Due / Notes
Hands-on Lab 1	Week 1	Electrical measurements and membrane RC circuits	Meets Aug 23–27; report due Sun Sep 5
Dry-lab workshop	Week 2	RC data analysis and introduction to neural simulation	Meets Aug 30–Sep 3

Lab / Workshop	Week	Topic	Due / Notes
Hands-on Lab 2	Week 3	Saline electrodes and bioelectric recording interfaces	Meets Sep 7–10; report due Sun Sep 19
Dry-lab workshop	Week 4	Electrode measurements, uncertainty, and report development	Meets Sep 13–17
Hands-on Lab 3	Week 5	Electronic neurons: threshold and firing	Meets Sep 20–24; report due Sun Oct 3
Dry-lab workshop	Week 6	Ion-channel dynamics and action-potential simulation	Meets Sep 27–Oct 1
Dry-lab workshop	Week 7	Cable and synaptic models; meetings before Fall Recess only	Meets Oct 4–6; no meetings Oct 7–8
Hands-on Lab 4	Week 8	Analog dendritic cables and synaptic integration	Meets Oct 11–15; report due Sun Oct 24
Hands-on Lab 5	Week 9	Surface EMG and muscle recruitment	Meets Oct 18–22; report due Sun Oct 31
Dry-lab workshop	Week 10	EMG analysis, model comparison, and simulation development	Meets Oct 25–29
Hands-on Lab 6	Week 11	Force sensing and closed-loop motor control	Meets Nov 1–5; report due Sun Nov 14
Hands-on Lab 7	Week 12	Acoustic sensing and sound localization	Meets Nov 8–12; report due Sun Nov 21
Hands-on Lab 8	Week 13	Photodiode sensors and visual-system circuit prototypes	Meets Nov 15–19; report due Fri Dec 3
Make-up laboratory only	Week 14	One missed hands-on experiment may be completed	Tue Nov 23; associated report due Fri Dec 3
Dry-lab workshop	Week 15	Simulation demonstrations and final laboratory workshop	Meets Nov 29–Dec 3
Workshop checkpoint 1	Fri Sep 10	RC measurements and model fitting	Due 11:59 PM
Workshop checkpoint 2	Fri Oct 1	Ion-channel and action-potential simulation	Due 11:59 PM
Workshop checkpoint 3	Fri Oct 29	EMG analysis and model comparison	Due 11:59 PM
Workshop checkpoint 4	Fri Nov 19	Model validation against experimental measurements	Due 11:59 PM

Lab sections meet weekly at the times and locations listed under Course Times & Locations. No meetings take place on university holidays; workshops affected by Fall Recess follow the announced adjustment, while sections affected by Veterans Day complete the required experiment in an alternate session outside the holiday. Week 14 contains only the November 23 make-up session. All homework, report, project, and workshop deadlines are 11:59 PM Pacific time.

Lecture Recordings and Study Materials

Lectures are recorded. Recordings are released **one week before the exam that covers them**. I may release them two weeks before an exam at my discretion, but early release is not guaranteed and should not be expected. You are responsible for attending lecture, taking thorough notes, and studying throughout the semester. **Do not wait for recordings to begin preparing for an exam.**

Recordings from previous offerings may also be posted as supplemental study materials. This course changes between semesters, and older recordings may omit current topics or include material that is no longer covered. **This semester's lectures determine what is examinable.** You are responsible for identifying those differences using the current lectures and slides.

Recordings do not replace attendance or excuse arriving unprepared. Lecture content is dense and requires consistent engagement. Choosing to rely on recordings does not reduce the material you are expected to know or change the standards by which your work will be graded.

Study Expectations

This course covers substantial biological and engineering material on a fast schedule. Review each lecture within 48 hours and work through its examples until you can explain why the method applies and what the answer means. Plan regular time for practice problems, model development, and laboratory analysis, and begin reports and projects early enough to identify questions before their deadlines.

Lecture & Lab Attendance

Lecture attendance is expected but does not carry a separate grade. Laboratory attendance is worth 10% of the laboratory component and includes both hands-on experiments and dry-lab workshops. One absence is excused without an attendance penalty; each additional unexcused absence reduces the overall laboratory grade by 5 percentage points. Contact your assigned TA about missed work and me directly about extensions, exceptional circumstances, or grading concerns.

Late Work

- Homework sets, lab reports, and assessed workshop work: 5% deduction from the available points per calendar day late for the first 7 days. Work more than one week late is not accepted without an approved extension. The approved make-up experiment follows its December 3 report deadline.
- **Missed laboratory work: The November 23 make-up day allows completion of one missed hands-on experiment. Further missed experiments require an individually approved arrangement when required by documented circumstances or university policy; they cannot be replaced with invented or copied measurements.**
- **Simulation project: 30% deduction for any submission after 11:59 PM on the due date. Projects submitted more than 48 hours late are not accepted.**

Extensions require instructor approval. Contact me before the deadline when possible, or as soon as reasonably possible in an emergency. Approved accommodations, religious

observance, and university-required arrangements take precedence over ordinary late penalties.

Academic Integrity and Use of AI Tools

Follow the university's standards of academic integrity. Cheating, fabricated measurements or citations, plagiarism, and unauthorized collaboration are handled under university procedures. Generative AI may not write lab report prose, individual scientific interpretations, or exam answers. Coding or troubleshooting assistance is permitted only when authorized and disclosed; verify the output and retain your original work. Shared apparatus and data collection are permitted where specified, while each report and the simulation project must reflect the student's own analysis and understanding. Do not upload classmates' recording data to external AI services.

Accommodations

Students seeking accommodations should contact the Office of Student Accessibility Services as early as possible. Provide the approved notice through the university process and coordinate lecture, lab, software, deadline, and exam arrangements with the instructor. Early notice supports planning; a course deadline does not cancel an approved accommodation. OSAS: https://osas.usc.edu/.

Communication

Questions about neural mechanisms, calculations, models, and measurements are expected. Your assigned TA is the first contact for homework, reports, software, and routine lab logistics. Contact me directly at jlopez@usc.edu for grading concerns and regrade requests, unresolved questions, accommodations, or confidential issues.

I will make every effort to respond to emails and messages promptly; however, because I teach multiple courses and run an active research laboratory, response times may be delayed. Students should generally expect responses within one business day. Messages sent in the evening, overnight, or on weekends will typically be addressed the following business day. Use clear subject lines that include the course number and topic, such as "BME 202: Lab 3 report question." Keep questions concise and include relevant context, such as the assignment name or lab section. Reach out early rather than waiting until deadlines or exam days. Consult the syllabus and current lecture materials for exam coverage before emailing.

Absences and Communication

I understand that many students balance academic responsibilities with work, family obligations, and other commitments. If you anticipate an absence or experience an emergency, contact me as soon as possible. For lab absences, notify your TA promptly and arrange the appropriate documentation, make-up, or equivalent task. Communicating early helps us find an appropriate arrangement.

Lecture recordings support review but do not replace laboratory attendance or the experimental work required for reports. The laboratory attendance policy applies to scheduled workshops as well as hands-on experiments, and university closures and approved university-

required arrangements do not count as unexcused absences. An excused absence does not automatically remove the requirement to complete graded work.

Course Calendar

Lectures meet Monday, Wednesday, and Friday for 80 minutes, beginning August 23 and ending December 3. Each of the 41 lecture meetings introduces a new topic through explanation and instructor-led worked examples. Exams 1–3 use the separate Monday session, 6:00–7:20 PM, on September 20, October 18, and November 8. Exam 4 is December 8 at 4:00 PM, the day after university study days. Regular lectures meet on exam days. Dates follow USC's Fall 2027 academic calendar (https://academics.usc.edu/calendar/). Assignment deadlines are 11:59 PM Pacific time unless otherwise stated.

Date (2027)	Session	Topic	Assessment / Notes
Mon Aug 23	Lab 1 · Week 1	Electrical measurements and membrane RC circuits	Meets Aug 23–27
Mon Aug 23	Lecture 1	The nervous system as a biomedical engineering system	
Wed Aug 25	Lecture 2	Measuring neural signals: voltage, current, and recording methods	
Fri Aug 27	Lecture 3	Ion channels: selectivity, conductance, and current flow	
Fri Aug 27	Homework 1 due	Problems on current lecture topics	11:59 PM
Mon Aug 30	Workshop · Week 2	RC data analysis and introduction to neural simulation	Meets Aug 30–Sep 3
Mon Aug 30	Lecture 4	Ion gradients and equilibrium potentials: the Nernst equation	
Wed Sep 1	Lecture 5	Resting membrane potential and the Goldman–Hodgkin–Katz equation	
Fri Sep 3	Lecture 6	Equivalent circuits of the neuronal membrane	
Fri Sep 3	Homework 2 due	Problems on current lecture topics	11:59 PM
Sun Sep 5	Lab 1 report due	Electrical measurements and membrane RC circuits	11:59 PM
Mon Sep 6	Holiday	Labor Day	No class or labs
Tue Sep 7	Lab 2 · Week 3	Saline electrodes and bioelectric recording interfaces	Meets Sep 7–10
Wed Sep 8	Lecture 7	Membrane capacitance and the neuronal RC circuit	
Fri Sep 10	Lecture 8	Passive membrane responses and the membrane time constant	
Fri Sep 10	Workshop checkpoint 1 due	RC measurements and model fitting	11:59 PM
Mon Sep 13	Workshop · Week 4	Electrode measurements, uncertainty, and report development	Meets Sep 13–17
Mon Sep 13	Lecture 9	Numerical simulation of membrane voltage: Euler integration	
Wed Sep 15	Lecture 10	Estimating membrane properties and testing a passive neuron model	
Fri Sep 17	Lecture 11	Voltage clamp: separating ionic currents and estimating conductance	

Date (2027)	Session	Topic	Assessment / Notes
Fri Sep 17	Homework 3 due	Problems on current lecture topics	11:59 PM
Sun Sep 19	Lab 2 report due	Saline electrodes and bioelectric recording interfaces	11:59 PM
Mon Sep 20	Lab 3 · Week 5	Electronic neurons: threshold and firing	Meets Sep 20–24
Mon Sep 20	Lecture 12	Ion-channel gating: opening probability and kinetic models	
Mon Sep 20	EXAM 1	Lectures 1–10	6:00–7:20 PM
Wed Sep 22	Lecture 13	The Hodgkin–Huxley model: voltage-dependent conductances	
Fri Sep 24	Lecture 14	Action potentials: threshold, refractoriness, and firing behavior	
Fri Sep 24	Homework 4 due	Problems on current lecture topics	11:59 PM
Fri Sep 24	Simulation proposal due	Neural system, model question, and initial evidence	11:59 PM
Mon Sep 27	Workshop · Week 6	Ion-channel dynamics and action-potential simulation	Meets Sep 27–Oct 1
Mon Sep 27	Lecture 15	Cable theory and passive spread through dendrites	
Wed Sep 29	Lecture 16	Action-potential propagation, myelination, and conduction velocity	
Fri Oct 1	Lecture 17	Neurotransmitter release and the reliability of synaptic transmission	
Fri Oct 1	Homework 5 due	Problems on current lecture topics	11:59 PM
Fri Oct 1	Workshop checkpoint 2 due	Ion-channel and action-potential simulation	11:59 PM
Sun Oct 3	Lab 3 report due	Electronic neurons: threshold and firing	11:59 PM
Mon Oct 4	Workshop · Week 7	Cable and synaptic models; meetings before Fall Recess only	Meets Oct 4–6
Mon Oct 4	Lecture 18	Postsynaptic currents: excitation, inhibition, and reversal potentials	
Wed Oct 6	Lecture 19	Dendritic integration of spatial and temporal inputs	
Thu Oct 7	Holiday	Fall Recess: October 7–8	No class or labs
Mon Oct 11	Lab 4 · Week 8	Analog dendritic cables and synaptic integration	Meets Oct 11–15
Mon Oct 11	Lecture 20	Synaptic plasticity: how activity changes communication	
Wed Oct 13	Lecture 21	Neural coding through firing rates and spike timing	
Fri Oct 15	Lecture 22	Population coding, tuning curves, and basic decoding	
Fri Oct 15	Homework 6 due	Problems on current lecture topics	11:59 PM
Mon Oct 18	Lab 5 · Week 9	Surface EMG and muscle recruitment	Meets Oct 18–22

Date (2027)	Session	Topic	Assessment / Notes
Mon Oct 18	Lecture 23	Neural filtering and system identification	
Mon Oct 18	EXAM 2	Lectures 11–20	6:00–7:20 PM
Wed Oct 20	Lecture 24	Excitation–inhibition balance and circuit dynamics	
Fri Oct 22	Lecture 25	Recurrent circuits, persistent activity, and memory	
Fri Oct 22	Homework 7 due	Problems on current lecture topics	11:59 PM
Fri Oct 22	Simulation progress due	Preliminary model and data comparison	11:59 PM
Sun Oct 24	Lab 4 report due	Analog dendritic cables and synaptic integration	11:59 PM
Mon Oct 25	Workshop · Week 10	EMG analysis, model comparison, and simulation development	Meets Oct 25–29
Mon Oct 25	Lecture 26	Feedback and feedforward control in neural systems	
Wed Oct 27	Lecture 27	Neuromuscular transmission, motor units, and muscle force	
Fri Oct 29	Lecture 28	Spinal reflexes and proprioceptive feedback	
Fri Oct 29	Homework 8 due	Problems on current lecture topics	11:59 PM
Fri Oct 29	Workshop checkpoint 3 due	EMG analysis and model comparison	11:59 PM
Sun Oct 31	Lab 5 report due	Surface EMG and muscle recruitment	11:59 PM
Mon Nov 1	Lab 6 · Week 11	Force sensing and closed-loop motor control	Meets Nov 1–5
Mon Nov 1	Lecture 29	Voluntary movement, sensorimotor adaptation, and error correction	
Wed Nov 3	Lecture 30	Circuit instability and the engineering of brain stimulation	
Fri Nov 5	Lecture 31	Somatosensory transduction and tactile encoding	
Fri Nov 5	Homework 9 due	Problems on current lecture topics	11:59 PM
Mon Nov 8	Lab 7 · Week 12	Acoustic sensing and sound localization	Meets Nov 8–12
Mon Nov 8	Lecture 32	Pain pathways and electrical approaches to pain modulation	
Mon Nov 8	EXAM 3	Lectures 21–30	6:00–7:20 PM
Wed Nov 10	Lecture 33	Cochlear mechanics, hair-cell transduction, and auditory-nerve coding	
Thu Nov 11	Holiday	Veterans Day	No class or labs
Fri Nov 12	Lecture 34	Auditory brainstem circuits and sound localization	
Sun Nov 14	Lab 6 report due	Force sensing and closed-loop motor control	11:59 PM
Mon Nov 15	Lab 8 · Week 13	Photodiode sensors and visual-system circuit prototypes	Meets Nov 15–19
Mon Nov 15	Lecture 35	Auditory cortex and the processing of complex sounds	

Date (2027)	Session	Topic	Assessment / Notes
Wed Nov 17	Lecture 36	Vestibular sensing, balance, and gaze stabilization	
Fri Nov 19	Lecture 37	Retinal transduction and visual cortical processing	
Fri Nov 19	Homework 10 due	Problems on current lecture topics	11:59 PM
Fri Nov 19	Workshop checkpoint 4 due	Model validation against experimental measurements	11:59 PM
Sun Nov 21	Lab 7 report due	Acoustic sensing and sound localization	11:59 PM
Mon Nov 22	Lecture 38	Neurotechnology: designing systems to measure and influence neural activity	
Tue Nov 23	Make-up laboratory	Complete only one missed hands-on experiment	Time announced
Wed Nov 24	Holiday	Thanksgiving: November 24–28	No class or labs
Mon Nov 29	Workshop · Week 15	Simulation demonstrations and final laboratory workshop	Meets Nov 29–Dec 3
Mon Nov 29	Lecture 39	Sensory prostheses: cochlear and retinal implants	
Wed Dec 1	Lecture 40	Neural electrodes: recording, stimulation, and tissue interfaces	
Fri Dec 3	Lecture 41	Brain–computer interfaces and closed-loop prosthetic control	
Fri Dec 3	Lab 8 report due	Photodiode sensors and visual-system circuit prototypes	11:59 PM
Fri Dec 3	Homework 11 due	Problems on current lecture topics	11:59 PM
Fri Dec 3	Final simulation due	Runnable model, figures, and code/notebook explanations	11:59 PM
Fri Dec 3	Make-up report due	Report for the approved November 23 experiment	11:59 PM
Sat Dec 4	Study days	December 4–7	No regular lectures or labs
Wed Dec 8	EXAM 4	Lectures 31–41	4:00 PM; room and duration TBA

Readings Calendar

Specific textbook selections and research-article readings will be added separately. Kandel and colleagues' Principles of Neural Science, 6th edition, is the optional biological reference; current lecture material and assigned readings determine exam preparation.

Final Word

This course asks you to understand the nervous system as both a biological system and an engineering problem. The goal is to explain how a mechanism works, predict its behavior, and use measurements to decide whether that explanation is adequate. By connecting lectures, worked examples, and laboratory experiments, you will develop a foundation for further study in biomedical engineering and neuroscience.

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